

Weekly Update 3

CSE299.1 Group - 3

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1. Backend Development (FastAPI):

- a. Developed a backend with **FastAPI**, creating multiple RESTful endpoints for handling user queries.
- b. Tested and validated the backend functionality using **ThunderClient** in Visual Studio Code, ensuring smooth communication between the client and server.

2. Document Embedding with FAISS:

- a. Integrated **FAISS** (Facebook AI Similarity Search) to handle document embedding and improve the chatbot's ability to retrieve relevant information quickly.
- b. The use of FAISS enhanced the chatbot's capacity to manage large datasets and improve response accuracy.

3. Frontend Development with Streamlit:

- a. Contributed to the development of the **Streamlit** frontend, creating a user-friendly interface for seamless chatbot interaction.
- b. Streamlit was used to design an intuitive layout, improving the overall user experience and ease of access.

4. Model Integration (Ollama):

- a. Integrated **Ollama models** to generate dynamic and context-aware responses for the chatbot.
- b. Faced challenges with implementing Ollama models on local devices, leading to a shift to **Google Colab** for a more efficient solution to resource limitations.

Challenges:

- **Local Device Limitations:** Significant obstacles were encountered in deploying Ollama models on local devices, causing performance issues. The transition to **Google Colab** allowed the project to overcome these limitations and improve model execution.

Next Steps:

1. Investigate and resolve the issues related to running Ollama models locally, ensuring a more efficient implementation for scalability.
2. Continue enhancing the **Streamlit** frontend to provide a more intuitive and interactive user experience.
3. Optimize **FAISS indexing** to reduce retrieval times and improve the overall performance of the chatbot.
4. Conduct comprehensive testing to refine the accuracy of the chatbot's responses and ensure overall system reliability.